Karthik Balasubramaniam

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Profile

Ph.D. astrophysicist specialising in multi-mission X-ray spectroscopy of active galactic nuclei, supermassive black holes and radio galaxies (Chandra, XMM-Newton, NuSTAR; XSPEC/relxill). Nine peer-reviewed publications and two first-author manuscripts in near-final preparation. Eight years of international research across India, Poland and Taiwan. Now returning to full-time research from Bengaluru, India; the intervening period produced applied data-pipeline and quality-engineering experience that directly reinforces reproducible scientific computing workflows. Strong in statistical inference, large multi-mission datasets and end-to-end analysis pipelines.

Research Experience

RAD@home Astronomy Collaboratory — **Bengaluru, India (remote)**
Remote Collaborator — *Jul 2026 – Present*
Invited X-ray collaborator on the collaboration's Collimated Synchrotron Thread (CST) radio-galaxy program (PI Prof. A. Hota): archival Chandra analysis of the CST-2 cluster environment (Abell 2092; $kT \approx 2.7$ keV) and a radio/X-ray pressure-balance test of the external confinement of the radio thread.

X-ray Astrophysics — **Remote**
Independent Researcher — *Aug 2023 – Present*
Two concurrent first-author projects:

- *Centaurus A giant-lobe X-ray point-source census* (185 sources; Chandra, full reproducible CIAO 4.18 pipeline; target **A&A**, in prep).
- *Supermassive black-hole spin via relativistic reflection* (relxill; multi-mission XMM/NuSTAR; target **ApJ**, in prep).

National Tsing Hua University — **Hsinchu, Taiwan**
Postdoctoral Research Fellow — *Aug 2022 – Jul 2023*
Multi-terabyte X-ray dataset analysis; reproducible Python/CIAO/XSPEC workflows; spectral modelling under Dr. Hsiang-Kuang Chang; 10+ international talks.

Astronomical Observatory, Jagiellonian University — **Kraków, Poland**
Doctoral Researcher — *Oct 2016 – Oct 2021*
Thesis: *Fluorescent Iron Lines in Various Types of Radio-Loud Active Galactic Nuclei*, under Prof. Łukasz Stawarz (co-advisor Dr. Volodymyr Marchenko). Chandra/XMM/NuSTAR spectroscopy & imaging; jet–ISM interactions; image deconvolution. Sources: Centaurus A, Pictor A, NGC 3894, CGCG 292-057, 4C+74.26.

Indian Institute of Astrophysics — **Bengaluru, India**
M.Sc. Research — *2014 – 2016*
Search for an unidentified 3.5 keV emission line in stacked X-ray spectra of dwarf galaxies (dark-matter candidates), with Dr. F. K. Sutaria & Dr. Subinoy Das.

Industry & Professional Experience

Ppyrus India Pvt. Ltd. — **Bengaluru, India**
Innovation & Ecosystem Associate — *May 2026 – Present*
Coordinate academic institutions and technology partners on digital-platform initiatives; data validation, metadata management, structured data integration and pipeline quality-assurance across multidisciplinary projects — directly transferable skills applied to reproducible scientific workflows.

Wistron Corporation — **Hsinchu, Taiwan**
Technical Assistant — *Nov 2025 – Feb 2026*
Semiconductor / PCB defect analysis, root-cause troubleshooting and test-protocol development; built defect-traceability documentation and QC validation workflows.

Vacatronics Technologies **Taiwan**
Maintenance Associate — Cleanroom *Mar 2024 – May 2025*
Class-100 cleanroom maintenance for optical-coating production: routine facility upkeep and replacement of parts on laser and vacuum coating chambers; process documentation and cross-team coordination.

HCL Technologies **Poland**
Senior Data Analyst *Jan 2022 – May 2022*
Metadata quality and catalogue consistency across 10,000+ product listings; SQL-based data validation, de-duplication and classification.

Teaching, Mentoring & Outreach

Cram Schools (buxiban), Taiwan
STEM & English Instructor *2023 – 2026*
Taught mathematics, science and English at **American Eagle** (Hsinchu), **XL Academy** (Taipei) and **Little Kangaroo** (Taipei); curriculum design and individualised instruction for diverse age groups.

National Tsing Hua University **Hsinchu, Taiwan**
Workshop Instructor *2022 – 2023*
Python-based astronomical data analysis and CIAO/XSPEC spectral fitting; trained 50+ researchers in multi-mission X-ray techniques.

Jagiellonian University **Kraków, Poland**
Mentor & Teaching Assistant *2016 – 2021*
Mentored 4+ students in high-energy astrophysics and computational methods.

Jagiellonian Observatory (OAUJ)
Science Outreach & Web Developer *2018 – 2022*
Built and maintained the "Night under the Stars" (NuTS) outreach website from scratch (HTML/CSS/JS). karthikab7.github.io/nuts

Entrepreneurship

Restaurant Venture **Taiwan**
Investor & Business Partner *2024 – Mar 2026*
Co-invested in a hospitality / restaurant business; involved in capital planning and growth strategy.

Education

Jagiellonian University **Kraków, Poland**
Ph.D., Astrophysics *2016 – 2021*
Defended Oct 2021. Advisor: Prof. Łukasz Stawarz (Department Head).

Christ University **Bengaluru, India**
M.Sc. Physics (Astrophysics), GPA 7.53 *2014 – 2016*

Visvesvaraya Technological University **India**
B.E. Mechanical Engineering, First Class *2009 – 2013*

Technical Skills

X-ray analysis: XSPEC, relxill, CIAO (Chandra), XMM-SAS, NuSTAR, HEASoft, fermiPy, AIPS, IRAF
Programming: Python (NumPy, SciPy, Astropy, Pandas, Matplotlib), C/C++, SQL, Bash
Computing: Linux/Unix, Git, Anaconda, Jupyter, reproducible pipelines, MCMC, HPC
Plotting / CAD: TOPCAT, Gnuplot, Origin; SolidWorks, ANSYS, Pro-E
Web & docs: HTML/CSS, JavaScript, LaTeX

Professional Development (2026)

- be10x — AI Tools & ChatGPT for technical productivity (2026).

- Generative AI & Large Language Models — applied prompt engineering.
- Advanced Python for Data Science & Machine Learning.
- SQL & modern data-analytics workflows.

Scholarships, Funding & Patents

- Overseas Fellowship DSTW/ED-1/CR-4/2016-17, Government of Karnataka, India (2016–2021).
- Polish National Science Centre (NCN) Grant 2016/22/E/ST9/00061 (PI: Ł. Stawarz).
- VSIP Fellowship, Indian Institute of Astrophysics (2014).
- Patents: US11668926B2; Taiwan TW109125878, TW110124889 (optical light-guide design).

Selected Publications

9 peer-reviewed (7 journal + 2 IAU proceedings) | 66 citations, h-index 4 (Google Scholar, Sep 2026) | 2 first-author manuscripts in preparation.

- E. Bronzini, ... **K. Balasubramaniam** et al., "Investigating X-ray emission in the GeV-emitting CSOs PKS 1718-649 & TXS 1146+596", *A&A* 684, A65 (2024).
- **K. Balasubramaniam** et al., "X-ray Emission of the γ-ray-Loud Young Radio Galaxy NGC 3894", *ApJ* 922, 84 (2021).
- R. Thimmappa, ... **K. Balasubramaniam** et al., "Complex Structure of the Eastern Lobe of Pictor A", *ApJ* 921, 44 (2021).
- **K. Balasubramaniam** et al., "Chandra View of the LINER-type Nucleus in CGCG 292-057: Ionized Iron Line & Jet–ISM Interactions", *ApJ* 905, 148 (2020).
- R. Thimmappa, ... **K. Balasubramaniam** et al., "Chandra Imaging of the Western Hotspot in Pictor A", *ApJ* 903, 109 (2020).
- G. Bhatta, Ł. Stawarz, A. Markowitz, **K. Balasubramaniam** et al., "Disk–Jet Coupling in the Broad-line Radio Quasar 4C+74.26", *ApJ* 866, 132 (2018).
- P. Deshmukh, P. Parihar, **K. Balasubramaniam** et al., "Dynamic Loading Assembly for Testing Actuators of a Segmented Mirror Telescope", *J. Astron. Instrum.* 6 (2017).
- **K. Balasubramaniam** et al., "Chandra view on the active nucleus of CGCG 292-057: jet–ISM interactions", *Proc. IAU* 342, 222 (2019).
- R. Thimmappa, ... **K. Balasubramaniam** et al., "Preliminary analysis of the X-ray emission from the central regions of Pictor A", *Proc. IAU* 342, 224 (2019).

In prep.: Centaurus A giant-lobe X-ray census (*A&A*); SMBH spin via relativistic reflection (*ApJ*).

Selected Talks & Conferences

- "Searching the Fe line with nearby AGN using Chandra data", Physical Society of Taiwan Annual Meeting (2023).
- "Fluorescent Iron Lines in Radio-Loud AGN", Astronomy Colloquium, NTHU (2022); Extreme Astro Group, U. Michigan (2022).
- Posters: "Black Hole Spins in AGNs: a systematic perspective" & "Exploring Centaurus-A's giant radio lobes", European Astronomical Society 2023, X-ray Universe 2023, ASROC 2023.
- "CSO NGC 3894 with restarting radio jets", 6th Workshop on CSS & GPS Radio Sources (2021).
- "Constraining the accretion-disk structure in 4C+74.26", Cracow School of Theoretical Physics, Zakopane (2019).
- IAU Symposium 342, Noto, Italy (2018); 20 Years of Chandra Symposium, Boston (2019); X-ray Universe, Rome (2017).